

ECEN 714

Lab3: Cell Characterization using Spectre

Name: Sheena Selvakumar

UIN:537000654

TA:Lilith Xu

Lab section:609

1. Opening the file: Cell18.spi file:

```
cell18.spi
~/ECEN-714/cellcharacs

//Spice netlist for an inverter
simulator lang=spectre
subckt IV (input output VDD VSS)
parameters wp=0.6u lp=0.2u wn=0.3u ln=0.3u
M1 output input VDD VDD tsmc18p w=wp l=lp
M2 output input VSS VSS tsmc18n w=wn l=ln
ends IV
```

Model18.spi:

```
model18.spi
~/ECEN-714/cellcharacs

* T77A SPICE BSIM3 VERSION 3.1 PARAMETERS
* SPICE 3f5 Level 8, Star-HSPICE Level 49, UTMOST Level 8
* DATE: Oct 17/07
* LOT: T77A WAF: 3010
* Temperature parameters=Default
.MODEL tsmc18n NMOS (
+VERSION = 3.1 TNOM = 27 TOX = 4.1E-9 LEVEL = 11
+XJ = 1E-7 NCH = 2.3549E17 VTH0 = 0.3729594
+K1 = 0.5840975 K2 = 1.686187E-3 K3 = 1E-3
+K3B = 0.0296594 W0 = 1E-7 NLX = 1.542817E-7
+DVT0W = 0 DVT1W = 0 DVT2W = 0
+DVT0 = 1.1680674 DVT1 = 0.4182499 DVT2 = 0.0268331
+U0 = 292.3966054 UA = -1.219746E-9 UB = 2.307456E-18
+UC = 6.953926E-11 VSAT = 1.704108E5 A0 = 1.0603725
+AGS = 0.4358079 B0 = 1.843620E-7 B1 = 5E-6
+KETA = -0.011523 A1 = 8.967934E-4 A2 = 0.3
+RDSW = 185.3073514 PRWG = 0.489299 PRWB = -0.2
+WR = 1 WINT = 0 LINT = 1.963253E-8
+XL = 0 XW = -1E-8 DWG = -5.54717E-9
+DWB = -1.072339E-8 VOFF = -0.0948017 NFACTOR = 2.1860065
+CTT = 0 CISC = 2.4E-4 CISC0 = 0
+CSICB = 0 ET40 = 1.925032E-3 ET4B = 6.028975E-5
+DSUB = 0.0193048 PCLM = 1.9022344 PDIBLC1 = 0.2550871
+PDIBLC2 = 1.417207E-3 PDIBLCB = -0.1 DROUT = 0.0645309
+PSCBE1 = 3.419362E10 PSCBE2 = 2.777738E-8 PVAG = 9.459570E-3
+DELTA = 0.01 RSH = 7 MOBMOD = 1
+PRT = 0 UTE = -1.5 KT1 = -0.11
+KT1L = 0 KT2 = 0.022 UA1 = 4.31E-9
+UB1 = -7.61E-18 UC1 = -5.6E-11 AT = 3.3E4
+WL = 0 WLN = 1 WW = 0
+WWN = 1 WML = 0 LWN = 1
+LLN = 1 LW = 0 LWN = 1
+LWL = 0 CAPMOD = 2 XPART = 0.5
+CGD0 = 8.58E-10 CGS0 = 8.58E-10 CGB0 = 1E-12
+CJ = 9.465842E-4 PB = 0.8 MJ = 0.3722711
```

Inverter.spi:

```
inverter.spi
~/ECEN-714/cellcharacs

:Spice netlist for an inverter and a capacitor
simulator lang=spectre

include "~/ecen454_714/cellcharacs/model18.spi"
include "~/ecen454_714/cellcharacs/cell18.spi"

vngd (gnd 0) vsource dc=0
vdd (vdd 0) vsource dc=1.8

vpwl (IV_in 0) vsource type=pwl wave={0n 0 1n 0 1.2n 1.8 4n 1.8 4.2n 0}

X1 (IV_in IV_out vdd gnd) IV wp=0.6u lp=0.2u wn=0.3u ln=0.2u

R1 (IV_out 1) resistor r=1
C1 (1 0) capacitor c=100f

TransientAnalysis tran start=0 stop=10ns step=1ps
save IV_in IV_out
```

So like i faced some issues while running the following commands like:

source

/opt/coe/cadence/SPECTRE/setup.SPECTRE.linux.b
ash spectre inverter.spi

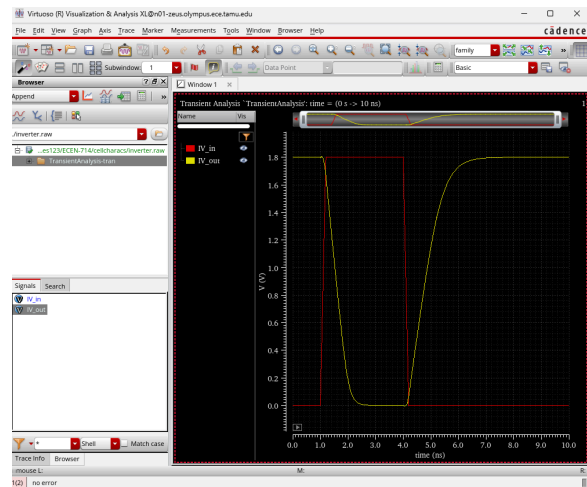
Turns out that my directory was different from what was present in the spice netlist, so i had to change that,

Here is the screenshot after i run the simulation:

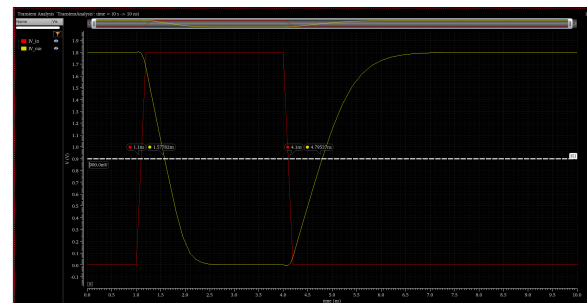
```
1:01:44:00.410000: VPIAuto_Hitl_0000_simulation

Aggregate audit (1:44:38 AM, Thur Oct 1, 2020):
Time used: CPU = 145 ms, elapsed = 230 ms, util. = 61.4%.
Time spent in licensing: elapsed = 33.1 ms, percentage of total = 14%.
Peak memory used = 107 Mbytes.
Simulation started at: 1:44:38 AM, Thur Oct 1, 2020, with elapsed time (wall clock): 236 ms.
spectre completes with 0 errors, 2 warnings, and 8 notices.
[shes22@n01-zeus cellcharacs]$
```

After running the cadence spectre, the following window appeared:



After placing the Markers:



1. Extracted Marker Values (at 50% Threshold = 0.9 V)

- Input Rising Edge ($V_{in} = 0.9V$):

$$t_{in,rise} = 1.10000ns$$

- Output Falling Edge ($V_{out} = 0.9V$):

$$t_{out,fall} = 1.57782ns$$

- **Input Falling Edge ($V_{in} = 0.9V$):**
 $t_{in,fall} = 4.10000ns$
- **Output Rising Edge ($V_{out} = 0.9V$):**
 $t_{out,rise} = 4.79537ns$

2. Propagation Delay Calculations

- **Falling Propagation Delay (t_{pHL}):**

$$t_{pHL} = t_{out,fall} - t_{in,rise} = 1.57782ns - 1.10000ns = 0.47782ns$$

- **Rising Propagation Delay (t_{pLH}):**

$$t_{pLH} = t_{out,rise} - t_{in,fall} = 4.79537ns - 4.10000ns = 0.69537ns$$

3. Delay Error Percentage

$$\text{Error \%} = (|t_{pLH} - t_{pHL}| / \min(t_{pLH}, t_{pHL})) \times 100$$

$$\text{Error \%} = (|0.69537ns - 0.47782ns| / 0.47782ns) \times 100 = 45.53\%$$

Parameter	Measured Value	Unit
Falling Propagation Delay (t_{pHL})	0.4778	ns
Rising Propagation Delay (t_{pLH})	0.6954	ns
Delay Error Percentage	45.53%	%

Before Proceeding to the delay table just wanted to make sure that the error is within the 10% error ,so doing the simulation again.(Increasing the width of the pmos)

```
;Spice netlist for an inverter and a capacitor
simulator lang=spectre

include "~/ECEN-714/cellcharacs/model18.spi"
include "~/ECEN-714/cellcharacs/cell18.spi"

vgn0 (gnd 0) vsource dc=0
vdd0 (vdd 0) vsource dc=1.8

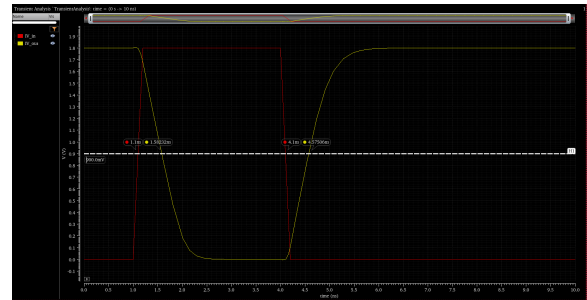
vpwl (IV_in 0) vsource type=pwl wave=[0n 0 1n 0 1.2n 1.8 4n 1.8 4.2n 0]

X1 (IV_in IV_out vdd gnd) IV wp=0.95u lp=0.2u wn=0.3u ln=0.2u

R1 (IV_out 1) resistor r=1
C1 (1 0) capacitor c=100f

TransientAnalysis tran start=0 stop=10ns step=1ps
save IV_in IV_out
```

Running the Simulation again;



1. Extracted Marker Values (at 50% Threshold = 0.9 V)

- **Input Rising Edge ($V_{in} = 0.9V$):**
 $t_{in,rise} = 1.10000ns$
- **Output Falling Edge ($V_{out} = 0.9V$):**
 $t_{out,fall} = 1.58232ns$
- **Input Falling Edge ($V_{in} = 0.9V$):**
 $t_{in,fall} = 4.10000ns$
- **Output Rising Edge ($V_{out} = 0.9V$):**
 $t_{out,rise} = 4.57506ns$

2. Propagation Delay Calculations

- **Falling Propagation Delay (t_{pHL}):**

$$t_{pHL} = t_{out,fall} - t_{in,rise} = 1.58232ns - 1.10000ns = 0.48232ns(48.23\%)$$

- **Rising Propagation Delay (t_{pLH}):**

$$t_{pLH} = t_{out,rise} - t_{in,fall} = 4.57506ns - 4.10000ns = 0.47506ns(47.50\%)$$

3. Delay Error Percentage

$$\text{Error \%} = (|t_{pLH} - t_{pHL}| / \min(t_{pLH}, t_{pHL})) \times 100$$

$$\text{Error \%} = (|0.47506 \text{ ns} - 0.48232 \text{ ns}| / 0.47506 \text{ ns}) \times 100 = (0.00726 / 0.47506) \times 100 \approx 1.53\%$$

Parameter	Measured Value	Unit
Falling Propagation Delay (t_{pHL})	0.4823	ns
Rising Propagation Delay (t_{pLH})	0.4751	ns
Delay Error Percentage	1.53%	%

Now creating and making the inverter delay table file starting from 1 farad:

```

;Spice netlist for an inverter and a capacitor
simulator lang=spectre

include "-/ECEN-714/cellcharacs/model18.spi"
include "-/ECEN-714/cellcharacs/cell18.spi"

vgnd (gnd 0) vsource dc=0
vvdv (vdd 0) vsource dc=1.8

vpwl (IV_in 0) vsource type=pwl wave=[0n 0 1n 0 1.2n 1.8 4n 1.8 4.2n 0]

X1 (IV_in IV_out vdd gnd) IV wp=0.95u lp=0.2u wn=0.3u ln=0.2u

;R1 (IV_out 1) resistor r=1
C1 (IV_out 0) capacitor c=1f

TransientAnalysis tran start=0 stop=10ns step=1ps
save IV_in IV_out
  
```

Computing the delay table based on different values of capacitances and computing the error voltage:

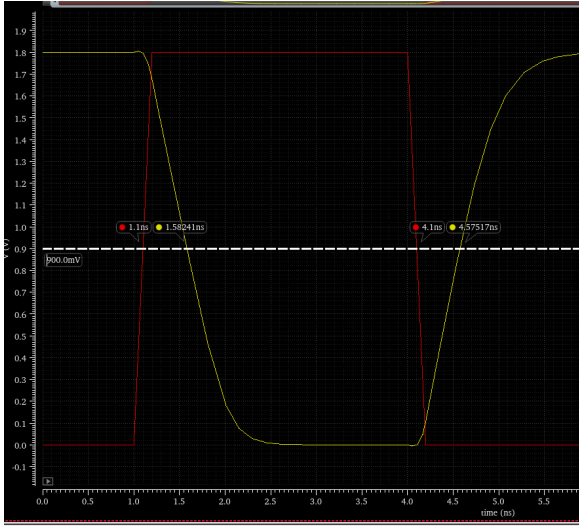
Load Capacitance (CL)	tpH (ns)	tpL (ns)	tp (ns)	Delay Error (%)

1 fF	0.0285	0.0441	0.0363	54.85%
8 fF	0.0759	0.0886	0.0822	16.72%
15 fF	0.1099	0.1187	0.1143	8.07%
22 fF	0.1405	0.1476	0.1441	5.07%
29 fF	0.1715	0.1771	0.1743	3.25%
36 fF	0.2023	0.2065	0.2044	2.11%
43 fF	0.2325	0.2346	0.223	0.91%

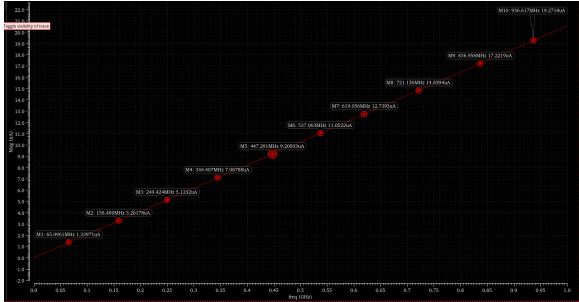
			3 6	
50 fF	0.2 629	0.2 656	0. 2 6 4 2	1.03%
64 fF	0.3 246	0.3 246	0. 3 2 4 6	0.00%
71 fF	0.3 566	0.3 557	0. 3 5 6 1	0.24%
78 fF	0.3 858	0.3 848	0. 3 8 5 3	0.25%
85 fF	0.4 172	0.4 125	0. 4 1 4 8	1.14%
92 fF	0.4 478	0.4 391	0. 4 4 3 4	1.99%

100 fF	0.4 824	0.4 752	0. 4 7 8 8	1.52%
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SCREENSHOT FOR THE 100F CAP:



Plotting the Sink Capacitance:



Calculation and tabulation:

Poi nt	Frequenc y (MHz)	Measured Current (μA)	Csin k (fF)
1	100	2.058	3.28
2	200	4.116	3.28

3	300	6.174	3.28
4	400	8.232	3.28
5	500	10.290	3.28
6	600	12.348	3.28
7	700	14.406	3.28
8	800	16.464	3.28
9	900	18.522	3.28
10	1000	20.580	3.28
Average	—	—	3.28 fF

Now repeating the same thing for **NAND**:
Including subcircuit information into the cell18.spi:

```

cell18.s
~/ECEN-714/cel

//Spice netlist for an inverter
simulator lang=spectre
subckt IV (input output VDD VSS)
    parameters wp=0.6u lp=0.2u wn=0.3u ln=0.3u
    M1 output input VDD VDD tsmc18P w=wp l=lp
    M2 output input VSS VSS tsmc18N w=wn l=ln
ends IV

subckt NAND2 (A B OUT VDD VSS)
    parameters wp=0.9u lp=0.2u wn=0.4u ln=0.2u
    // PMOS transistors in parallel
    M1 (OUT A VDD VDD) tsmc18P w=wp l=lp
    M2 (OUT B VDD VDD) tsmc18P w=wp l=lp
    // NMOS transistors in series
    M3 (OUT A net_n1 VSS) tsmc18N w=wn l=ln
    M4 (net_n1 B VSS VSS) tsmc18N w=wn l=ln
ends NAND2

```

Created the netlist and like updated the nand2 spi:

Nand 2 spice netlist:

```

nand2.spi
~/ECEN-714/cellcharacs

;Spice netlist for an inverter and a capacitor
simulator lang=spectre

include "~/ECEN-714/cellcharacs/model18.spi"
include "~/ECEN-714/cellcharacs/cell18.spi"

vgnd (gnd 0) vsourse dc=0
vvdv (vdd 0) vsourse dc=1.8

vpwl1 (IV_in1 0) vsourse type=pwl wave=[0n 0 1n 1.2n 1.8 4n 1.8 4.2n 0]
vpwl2 (IV_in2 0) vsourse dc=1.8

; Input A connected to PWL source (IV_in1), Input B connected to VDD (1.8V)
X1 (IV_in1 IV_in2 IV_out vdd gnd) NAND2 wp=0.95u lp=0.2u wn=0.5u ln=0.2u

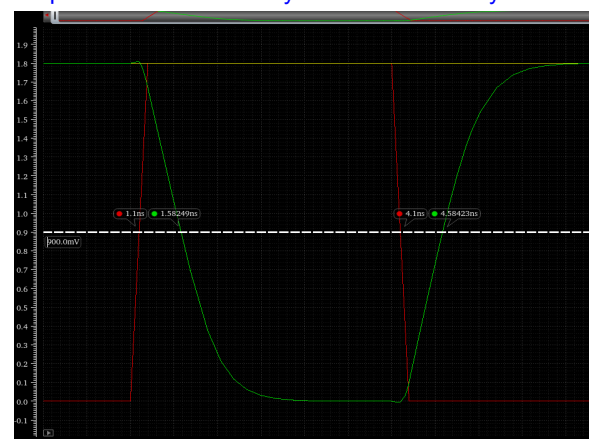
R1 (IV_out 1) resistor r=1
C1 (1 0) capacitor c=100f

TransientAnalysis tran start=0 stop=10ns step=1ps
save IV_in1 IV_in2 IV_out

```

Delay calculation:

Graph Obtained to actually calculate the delay:



Calculation of delay under 10%:

- **Falling Propagation Delay (t_{pHL}):**

$$t_{pHL} = t_{out,fall} - t_{in,rise} = 1.58249ns - 1.10000ns = 0.48249ns(482.49)$$

- **Rising Propagation Delay (t_{pLH}):**

$$t_{pLH} = t_{out, rise} - t_{in, fall} = 4.58423ns - 4.10000ns = 0.48423ns$$

- **Average Propagation Delay (t_p):**

$$t_p = \frac{t_{pHL} + t_{pLH}}{2} = \frac{0.48249ns + 0.48423ns}{2} = 0.48336ns(483.36ps)$$

$$\text{Error \%} = (|0.48423 \text{ ns} - 0.48249 \text{ ns}| / 0.48249 \text{ ns}) \times 100 = (0.00174 / 0.48249) \times 100 \approx 0.36\%$$

NAND DELAY FOR DELAY TABLE:WITHOUT RESISTOR AND CAPACITOR

```

Open  [icon] *nand2_delaytable.spi
~\ECEN-714\cellcharacs
;Spice netlist for an inverter and a capacitor
simulator lang=spectre

include "~\ECEN-714\cellcharacs\model18.spi"
include "~\ECEN-714\cellcharacs\cell18.spi"

vgnnd (gnd 0) vsource dc=0
vdd (vdd 0) vsource dc=1.8

vpwl1 (IV_in1 0) vsource type=pwl wave=[0n 0 1n 0 1.2n 1.8 4n 1.8 4.2n 0]
vpwl2 (IV_in2 0) vsource dc=1.8

; Input A connected to PWL source (IV_in), Input B connected to VDD (1.8V)
X1 (IV_in1 IV_in2 IV_out vdd gnd) NAND2 wp=0.95u lp=0.2u wn=0.5u ln=0.2u

;R1 (IV_out 1) resistor r=1
C1 (IV_out 0) capacitor c=1f

TransientAnalysis tran start=0 stop=10ns step=1ps
save IV_in1 IV_in2 IV_out

```

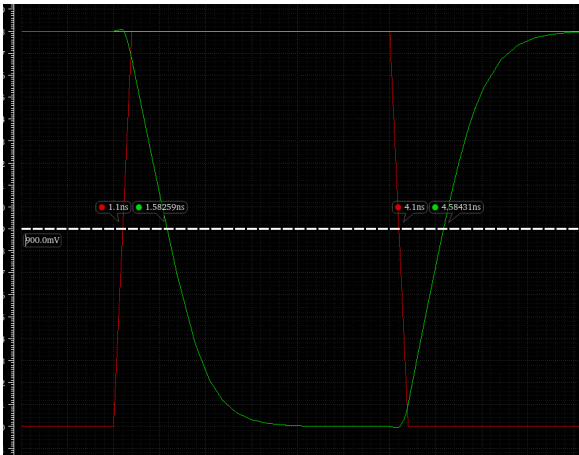
Table showing all the values:

Cloud (fF)	tpHL (ps)	tpLH (ps)	tp (ps)	Delay Error
1	34.57	63.69	49.13	84.23%
8	75.79	100.93	88.36	33.17%
15	109.06	130.00	119.53	19.20%

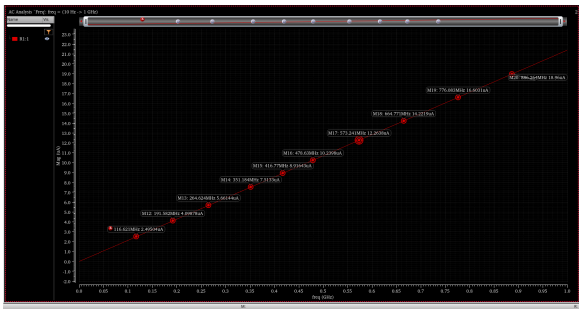
22	139.81	159.33	149.57	13.96%
29	170.64	188.63	179.64	10.54%
36	201.59	217.88	209.74	8.08%
43	232.17	247.36	239.77	6.54%
50	262.59	276.58	269.58	5.33%
57	294.80	305.17	299.99	3.52%
64	324.90	334.32	329.61	2.90%
71	354.85	363.73	359.29	2.50%
78	384.81	392.93	388.87	2.11%
85	417.23	422.13	419.68	1.17%

92	448.12	452.05	450.09	0.88%
100	482.59	484.31	483.45	0.36%

Graph showing the Plot for 100f



The AC Simulation Graph:



Marker Label	Frequency f (MHz)	Current I (μA)	Calculated Csink (fF)
A (M11)	116.621	2.49504	3.4050

M12	191.582	4.09878	3.4050
M13	264.624	5.66144	3.4050
M14	351.184	7.51330	3.4050
M15	416.770	8.91643	3.4050
M16	478.630	10.23980	3.4050
M17	573.241	12.26380	3.4049
M18	664.771	14.22190	3.4049
M19	776.083	16.60310	3.4049
M20	886.264	18.96000	3.4048

Average Sink Capacitance

$C_{sink,avg} = 3.4050fF$

Now Repeating the same thing for **xor** gate:
1.starting with the delay calculation:
Updating the file:

```
cell18.spi
~/ECEN-714/cellcharacs

//Spice netlist for an inverter
simulator lang=spectre
subckt IV (input output VDD VSS)
    parameters wp=0.6u lp=0.2u wn=0.3u ln=0.3u
    M1 output input VDD VDD tsmc18P w=wp l=lp
    M2 output input VSS VSS tsmc18N w=wn l=ln
ends IV

subckt NAND2 (A B OUT VDD VSS)
    parameters wp=0.9u lp=0.2u wn=0.4u ln=0.2u
    // PMOS transistors in parallel
    M1 (OUT A VDD VDD) tsmc18P w=wp l=lp
    M2 (OUT B VDD VDD) tsmc18P w=wp l=lp
    // NMOS transistors in series
    M3 (OUT B net_n1 VSS) tsmc18N w=wn l=ln
    M4 (net_n1 A VSS VSS) tsmc18N w=wn l=ln
ends NAND2

subckt XOR2 (A B OUT VDD VSS)
    parameters wp=0.9u lp=0.2u wn=0.4u ln=0.2u
    // Inverters for inputs A and B
    M1 (An A VDD VDD) tsmc18P w=wp l=lp
    M2 (An A VSS VSS) tsmc18N w=wn l=ln
    M3 (Bn B VDD VDD) tsmc18P w=wp l=lp
    M4 (Bn B VSS VSS) tsmc18N w=wn l=ln

    // Pull-up network
    M5 (net_p1 A VDD VDD) tsmc18P w=wp l=lp
    M6 (OUT Bn net_p1 VDD) tsmc18P w=wp l=lp
    M7 (net_p2 An VDD VDD) tsmc18P w=wp l=lp
    M8 (OUT B net_p2 VDD) tsmc18P w=wp l=lp

    // Pull-down network
    M9 (OUT A net_n1 VSS) tsmc18N w=wn l=ln
    M10 (net_n1 B VSS VSS) tsmc18N w=wn l=ln
    M11 (OUT An net_n2 VSS) tsmc18N w=wn l=ln
    M12 (net_n2 Bn VSS VSS) tsmc18N w=wn l=ln
ends XOR2
```

The xor spice file:

```
xor2.spi
~/ECEN-714/cellcharacs

; Spice netlist for XOR2 delay simulation
simulator lang=spectre

include "~/ECEN-714/cellcharacs/model18.spi"
include "~/ECEN-714/cellcharacs/cell18.spi"

vgnd (gnd 0) vsource dc=0
vdd (vdd 0) vsource dc=1.8

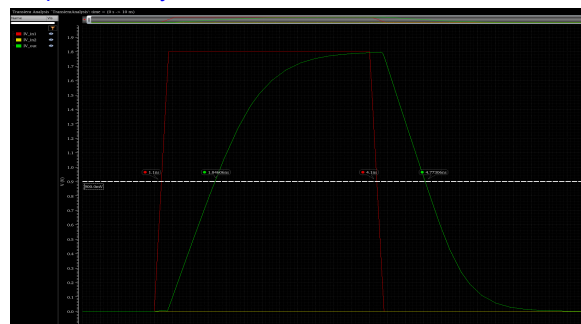
; Pulse input for Input A
vpwl1 (IV_in1 0) vsource type=pwl wave=[0n 0 1n 0 1.2n 1.8 4n 1.8 4.2n 0]
vpwl2 (IV_in2 0) vsource dc=0

; XOR2 Instance: Input A = IV_in1, Input B = vdd (1.8V), Output = IV_out
X1 (IV_in1 IV_in2 IV_out vdd gnd) XOR2 wp=1.2u lp=0.2u wn=0.4u ln=0.2u

R1 (IV_out 1) resistor r=1
C1 (1 0) capacitor c=100f

TransientAnalysis tran start=0 stop=10ns step=1ps
save IV_in1 IV_in2 IV_out
```

Graph for delay calculation:



- **Input Rising Edge ($V_{in1} = 0.9V$):**
 $t_{in1,rise} = 1.10000ns$
- **Output Rising Edge ($V_{out} = 0.9V$):**
 $t_{out,rise} = 1.84606ns$

- **Input Falling Edge ($V_{in1} = 0.9V$):**
 $t_{in1,fall} = 4.10000ns$
- **Output Falling Edge ($V_{out} = 0.9V$):**
 $t_{out,fall} = 4.77306ns$
- **Low-to-High Propagation Delay (t_{pLH}):**

$$t_{pLH} = 1.84606ns - 1.10000ns = 0.74606ns(746.06ps)$$

- **High-to-Low Propagation Delay (t_{pHL}):**

$$t_{pHL} = 4.77306ns - 4.10000ns = 0.67306ns(673.06ps)$$

- **Average Propagation Delay (t_p):**

$$t_p = \frac{746.06ps + 673.06ps}{2} = 709.56ps$$

- **Delay Error Percentage:**

$$\text{Error \%} = (|746.06 - 673.06| / 673.06) \times 100 = 10.85\%$$

Next is like the Delay Table for the xor gate:

The spice file required for the delay table:

```
xor2.spi
xor2_delaytable.spi

; Spice netlist for XOR2 delay simulation
simulator lang=spectre

include "~/ECEN-714/cellcharacs/model18.spi"
include "~/ECEN-714/cellcharacs/cell18.spi"

vgnd (gnd 0) vsource dc=0
vdd (vdd 0) vsource dc=1.8

; Pulse input for Input A
vpwl1 (IV_in1 0) vsource type=pwl wave=[0n 0 1n 0 1.2n 1.8 4n 1.8 4.2n 0]
vpwl2 (IV_in2 0) vsource dc=0

; XOR2 Instance: Input A = IV_in1, Input B = vdd (1.8V), Output = IV_out
X1 (IV_in1 IV_in2 IV_out vdd gnd) XOR2 wp=1.2u lp=0.2u wn=0.35u ln=0.2u

;R1 (IV_out 1) resistor r=1
C1 (IV_out 0) capacitor c=100f

TransientAnalysis tran start=0 stop=10ns step=1ps
save IV_in1 IV_in2 IV_out
```

The table for 15 values of capacitance:

Complete XOR Gate Propagation Delay vs. Load Capacitance Table

Clod (fF)	tpHL (ps)	tpLH (ps)	tp (ps)	Delay Error

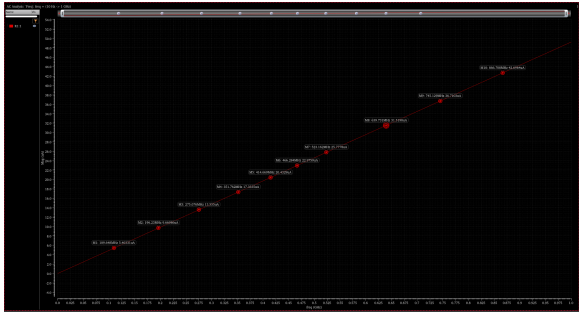
1	118.6 1	114.7 5	116 .68	3.36%
8	160.7 4	160.2 0	160 .47	0.34%
15	200.8 8	204.4 3	202 .66	1.77%
22	240.3 4	248.6 1	244 .48	3.44%
29	279.7 5	293.2 1	286 .48	4.81%

36	318.9 7	337.5 0	328 .24	5.81%
43	358.1 1	381.9 5	370 .03	6.66%
50	397.2 5	426.9 3	412 .09	7.47%
57	436.4 1	472.7 2	454 .57	8.32%
64	475.4 2	515.3 8	495 .40	8.41%

71	514.46	560.63	537.55	8.97%
78	553.50	604.28	578.89	9.17%
85	592.21	649.44	620.83	9.66%
92	630.19	693.11	661.65	9.98%

100	673.15	746.14	709.65	10.84%
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The AC Simulation:



The table required for the calculation of sink capacitance:

$$C_{sink} = \frac{I}{2\pi f}$$

XOR2 Sink Capacitance AC Simulation Table

Mar ker	Frequen cy f (MHz)	Current Magnitude I (μA)	Calculate d Csink (fF)
M1	109.648	5.40331	7.8429
M2	196.230	9.66986	7.8429
M3	275.076	13.55500	7.8427

M4	351.762	17.33350	7.8426
M5	414.669	20.43290	7.8424
M6	466.284	22.97590	7.8423
M7	523.162	25.77780	7.8421
M8	639.731	31.51980	7.8416
M9	745.129	36.71030	7.8411
M10	866.708	42.69640	7.8404

Average	—	—	7.8421
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Summary:
The Standard Logic Cells are been characterized by measuring their propagation delay and input capacitance through transistor level spice simulation